

2017 Ph H2 Q5

Section: Our Dynamic Universe

Topic: Measure, Gravitation, Relativity

A spacecraft moves at a constant speed past a stationary observer.

- Length of spacecraft in its own frame: $L_0 = 120\text{ m}$
- Observer measures contracted length: $L = 90\text{ m}$
- $c = 3.0 \times 10^8\text{ m/s}$

(a)

Calculate the speed of the spacecraft as measured by the observer.

Use the **length contraction** formula:

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Rearranged:

$$\frac{L}{L_0} = \sqrt{1 - \frac{v^2}{c^2}} \Rightarrow \left(\frac{L}{L_0}\right)^2 = 1 - \frac{v^2}{c^2}$$

Substitute:

$$\left(\frac{90}{120}\right)^2 = 1 - \frac{v^2}{(3.0 \times 10^8)^2} \Rightarrow (0.75)^2 = 1 - \frac{v^2}{9 \times 10^{16}} \Rightarrow 0.5625 = 1 - \frac{v^2}{9 \times 10^{16}}$$

$$\frac{v^2}{9 \times 10^{16}} = 0.4375 \Rightarrow v^2 = 3.9375 \times 10^{16} \Rightarrow v = \sqrt{3.9375 \times 10^{16}} = 1.98 \times 10^8\text{ m/s}$$

 *Revision tip:*

- Length contraction occurs **only in the direction of motion**
- Only moving observers see contracted lengths
- If length **shrinks**, you're in the **rest frame of the observer**